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LIQUID ROCKET SPRAY COMBUSTION STABILITY ANALYSIS

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Abstract

A computational methodology for the analysis of spray combustion stability in liquid rocket combustors has been constructed. The analysis is based on the unsteady, quasi-two-dimensional Euler equations with interphase source terms derived from a Lagrangian treatment of the combusting spray. An axisymetric combustor geometry is analyzed in terms of an equivalent annular computational domain with periodicity conditions which may further be specialized for a rectangular combustor geometry or reduced to a quasi-one-dimensional description for pure longitudinal-mode analyses. The propellant spray is represented by Lagrangian tracking of computational parcels each having a droplet spatial distribution described by a probability density function. The propellant burning rate is described by the d^2 vaporization law with the Ranz-Marshall convection correlation and a one-parameter phenomenological pressure-sensitive response index. The longitudinal- and tangential-mode instability characteristics of a generic combustor were investigated using the developed methodology. Intrinsic longitudinal-mode instability was observed at relatively high pressure-sensitive combustion response levels. The resulting limit cycle oscillation consisted of a single steep-fronted pressure wave undergoing repeated injector and nozzle reflections at the fundamental acoustical frequency. Intrinsic tangential-mode instability was observed even with vanishing pressure-sensitive combustion response. The observed instability exhibited mode switching from a single steep-fronted traveling pressure wave to two steep-fronted waves traveling in opposing directions. These extremely high-amplitude pressure waves were localized very near the injector face in a manner consistent with the shocklike waves observed in some experiments.

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Introduction

This paper reports on some recent efforts to apply modern computational techniques to the analysis of spray combustion stability in liquid-fueled rocket engines. Such efforts seem justified on the basis that numerical methods allow for a more comprehensive analysis of combustion instability than can be achieved using classical analytical treatments. By accounting for nonlinear behavior and allowing for more comprehensive physical submodeling, a computational approach becomes very alluring with the promise of shedding valuable insight on the mysterious connections between unsteady combustion and organized flow oscillations. From a practical standpoint, an efficient computational tool can also serve as a valuable design aid. In this paper, we outline our attempt to construct the basic framework for such an engineering analysis tool with emphasis on robustness and efficiency while capturing the essential features of the physical phenomena.

When attempting such a development, it is necessary to recognize that numerical solution techniques are beset with thier own special vexations in application and that physical theories for unsteady combustion phenomena retain a certain degree of uncertainty. Meaningful analysis of the problem is not yet a simple matter of mere implementation. For example, both the dissipative and dispersive characteristics of a particular scheme can have an important influence on numerical accuracy in unsteady flow problems as demonstrated by Hsieh.[1] Also, because combustion instability is basically a wave phenomenon, implementation of appropriate boundary conditions is not trivial. Stability is greatly determined by the transmission and reflection of flow disturbances incident on the boundaries. Furthermore, instability oscillation modes typically involve multidimensional effects and time accurate solutions are required imposing additional computational burden. The foremost difficulty, however, is the absence of definitive physical submodeling for unsteady combustion.

The complex physics of unsteady spray combustion in liquid rocket engines can still only be grossly modeled using our best contemporary descriptions. Even with these available descriptions, there is difficulty in implementing the modeling within a CFD framework without introducing excessive computational penalties or nonphysical numerical noise. For

instability problems, such nonphysical noise is of central importance. Strong numerical noise can falsely trigger nonlinear instabilities and yield misleading results.

The problems described above make the prediction and description of spray combustion instabilities especially difficult. Currently, stability analysis methods can serve only as guides leading to qualitative understanding rather than a priori quantitative predictions from first principles. By adapting contemporary computational techniques and refined physical submodeling, however, this situation may hopefully be improved. In fact, considerable progress is being made toward achieving this goal by several researchers.[2–8]

In this paper, our intent is to contribute to such an effort by developing a computational framework suitable for spray combustion stability analyses. Our desire is that the resulting development serve initially as a useful research code and ultimately as a design aide following the introduction of more complete physical submodeling. Emphasis is placed on computational efficiency and robustness. At this early stage of code development, a simple spray combustion physical submodel is considered with a one-parameter phenomenological pressure-sensitive combustion response index. It seems wise to first obtain results using a simplified instability model in lieu of more complicated physical submodeling with inherently more convoluted effects. With this controlled approach, we hope to develop a computational framework more efficiently and still gain insight into the basic physical phenomena. Once established, the retrogressive nature of this developmental approach may be relaxed by substituting more refined physical submodeling.

Theory

Physical Model

Recognizing that the physical model must ultimately describe transverse-mode instabilities but that computational efficiency is also essential, we have adapted the equivalent annular combustor representation for an actual axisymmetric combustor geometry as formulated by French researchers.[2,3] Although this approach neglects radial-mode instabilities, it allows for a description of the more important tangential-modes with the computational savings derived from using a two-dimensional grid. The resulting formulation may also be specialized for rectangular combustors or reduced to a quasi-one-dimensional description for pure longitudinal-mode analyses. The resulting computational framework is therefore quite versatile.

The geometry of an axisymmetric combustor and an equivalent annular computational domain is shown in Fig. 1. Here, the axial coordinate of the computational domain corresponds directly to that of the actual engine. The transverse length l, however, is chosen to match the desired n^{th} tangential mode of the real combustor. Periodicity of the transverse coordinate is also required. With these two lengths, the annular width b is fixed such that the cross-sectional areas match. Clearly, b is a function of axial position only. If analysis of a rectangular combustor is desired, l is taken as the height, b becomes a variable width, and the transverse periodicity condition is replaced by solid wall boundary conditions. For a quasi-one-dimensional analysis, l is taken as unity and b becomes a variable cross-sectional area.

We consider a bi-propellant system with injection of a gaseous oxidizer and a fuel spray at the injector face. Treatment of the gas phase was simplified by neglecting viscous/turbulence effects except for interphase momentum exchange. Physical modeling of the combustion process was simplified by invoking the following assumptions:

(1) Fuel is injected as a monodispersed spray of droplets – atomization is neglected.
(2) The local mean vaporization rate is determined by applying the quasi-steady d^2 vaporization law with corrections for convection and pressure sensitivity.
(3) Combustion is assumed to occur infinitely fast following vaporization of the fuel droplets.
(4) Analysis is restricted to fuel-lean operating conditions only so that all fuel is completely consumed.
(5) The gas-phase combustion products are assumed to be calorically as well as thermally perfect.

Gas-Phase Formulation

In our development, the gas phase is treated using an Eulerian description with appropriate source terms for interphase transport coupling with the spray. The problem thus reduces to solving the unsteady, quasi-two-dimensional Euler equations which may be written in conservative form using matrix notation as

$$\frac{\partial Q}{\partial t} + \frac{\partial E}{\partial x} + \frac{\partial F}{\partial y} = H \tag{1}$$

where the conserved variable vector Q is defined by

$$Q = \begin{bmatrix} \rho b \\ \rho u b \\ \rho v b \\ \rho e b \end{bmatrix} \tag{2}$$

and the convective flux vectors E and F are defined

by

$$E = \begin{bmatrix} \rho u b \\ (\rho u^2 + p)b \\ \rho u v b \\ (\rho e + p) u b \end{bmatrix} \quad F = \begin{bmatrix} \rho v b \\ \rho u v b \\ (\rho v^2 + p)b \\ (\rho e + p) v b \end{bmatrix} \tag{3}$$

The vector H contains contributions due to the variable annular width and interphase transport source terms. We denote this vector as the sum of two components

$$H = H_1 + H_2 \tag{4}$$

For single-phase flow, only H_1 need be considered to account for the variable annular width effect on the axial momentum equation. It consists of a single component

$$H_1 = \begin{bmatrix} 0 \\ p\frac{\partial b}{\partial x} \\ 0 \\ 0 \end{bmatrix} \tag{5}$$

For two-phase flow, the source vector H_2 must be included to account for interphase transport effects on the continuous-phase conservation equations. The source vector H_2 has the form

$$H_2 = \begin{bmatrix} b\Sigma\dot{w}_i \\ b\Sigma[\dot{w}_i u_{p_i} + \mathcal{D}_{x_i}] \\ b\Sigma[\dot{w}_i v_{p_i} + \mathcal{D}_{y_i}] \\ b\Sigma\dot{w}_i[\Delta q_i + \frac{q^2}{2}] + b\Sigma[u\mathcal{D}_{x_i} + v\mathcal{D}_{y_i}] \end{bmatrix} \tag{6}$$

The perfect gas equation of state $p = \rho RT = (\gamma - 1)[e - \rho\frac{q^2}{2}]$ completes the governing system where $e = C_v T + \frac{q^2}{2}$ represents the total internal energy.

This inhomogeneous form of the Euler equations retains a hyperbolic character. Thus, the Jacobian coefficient matrix $A = \partial E/\partial Q$ has real eigenvalues u, $u+c$, u, and $u-c$ with the speed of sound defined by $c = \sqrt{\gamma p/\rho}$. Similarly, the eigenvalues of the Jacobian coefficient matrix $B = \partial F/\partial Q$ are v, v, $v+c$, and $v-c$. Following conventional practice, A and B may be diagonalized such that

$$\begin{aligned} T_x^{-1} M^{-1} A M T_x &= \Lambda_x \\ T_y^{-1} M^{-1} B M T_y &= \Lambda_y \end{aligned} \tag{7}$$

where $M = \frac{\partial Q}{\partial V}$ is a transformation matrix from the conservative variables Q to the primitive variables V, T_x and T_y are similarity transformation matrices formed from the eigenvectors of the primitive variable Jacobian coefficient matrices $M^{-1}AM$ and $M^{-1}BM$, and Λ_x and Λ_y are the diagonal eigenvalue matrices.

For convienience, we may define positive and negative eigenvalue matrices as $\Lambda_x^{\pm}$ and $\Lambda_y^{\pm}$ where '+' indicates the occurence of positive eigenvalues only and '−' indicates only negative eigenvalues. It follows, therefore, that

$$\begin{aligned} A &= M T_x (\Lambda_x^+ + \Lambda_x^-) T_x^{-1} M^{-1} = A^+ + A^- \\ B &= M T_y (\Lambda_y^+ + \Lambda_y^-) T_y^{-1} M^{-1} = B^+ + B^- \end{aligned} \tag{8}$$

and

$$\begin{aligned} |A| &= M T_x (\Lambda_x^+ - \Lambda_x^-) T_x^{-1} M^{-1} = A^+ - A^- \\ |B| &= M T_y (\Lambda_y^+ - \Lambda_y^-) T_y^{-1} M^{-1} = B^+ - B^- \end{aligned} \tag{9}$$

Numerical Scheme

Numerical solution of the Euler equations is accomplished through optional spatial finite-difference discretizations and a separate time-integration method. In selecting discretization methods, we desired a conservative formulation capable of yielding meaningful spatial resolution without spurious numerical oscillations. In selecting a time-integration method, we desired meaningful time-accuracy and good numerical stability. These properties are central to the description of unsteady flows typically encountered in a combustion stability analysis. Therefore, both the dissipative and dispersive characteristics of a numerical scheme are important to the problem. Of course, efficiency is always of utmost concern.

An assessment of numerical techniques for such unsteady flow calculations has been carried out by Hsieh.[1] Based on his results, explicit Runge-Kutta multi-stage time-integration with 6th order central differencing of the convective fluxes and 4th order artificial dissipation yield very good all around performance. He also found that 2nd order, upwind flux-differencing with Roe averaging of the Jacobian coefficient matrices and nonlinear flux limiters to achieve TVD conditions performed very well with regard to shock capturing but entailed higher dissipative errors in multidimensional problems. Our construction of a computational method for spray combustion stability is based on adopting the best performing techniques from this study.

The development of the numerical scheme begins from the spatially discretized Euler equations yielding a system of ODEs of the form

$$\frac{dQ_{j,k}}{dt} = -R(Q_{j,k}) \tag{10}$$

where $R(Q_{j,k})$ represents an explicit spatial operator. Applying the 4-stage Runge-Kutta method yields the

operational sequence

$$
\begin{aligned}
Q_{j,k}^{(0)} &= Q_{j,k}^{n} \\
Q_{j,k}^{(1)} &= Q_{j,k}^{n} - \frac{1}{4}\Delta t R(Q_{j,k}^{(0)}) \\
Q_{j,k}^{(2)} &= Q_{j,k}^{n} - \frac{1}{3}\Delta t R(Q_{j,k}^{(1)}) \\
Q_{j,k}^{(3)} &= Q_{j,k}^{n} - \frac{1}{2}\Delta t R(Q_{j,k}^{(2)}) \\
Q_{j,k}^{n+1} &= Q_{j,k}^{n} - \Delta t R(Q_{j,k}^{(3)})
\end{aligned} \tag{11}
$$

With 6th order central differencing of the convective fluxes and the addition of 4th order artificial dissipation, the explicit spatial operator takes the form

$$
\begin{aligned}
R(Q_{j,k}) = \frac{1}{60\Delta x}\Big(& E_{j+3,k} - 9E_{j+2,k} + 45E_{j+1,k} \\
& -45E_{j-1,k} + 9E_{j-2,k} - E_{j-3,k}\Big) \\
+\frac{1}{60\Delta y}\Big(& F_{j,k+3} - 9F_{j,k+2} + 45F_{j,k+1} \\
& -45F_{j,k-1} + 9F_{j,k-2} - F_{j,k-3}\Big) \\
& +D_{j,k}^{(x)} + D_{j,k}^{(y)} - H_{j,k}
\end{aligned} \tag{12}
$$

The dissipation vector for the streamwise coordinate is[9]

$$
\begin{aligned}
D_{j,k}^{(x)} = \frac{\epsilon}{8\Delta t}\Big(& Q_{j+2,k} - 4Q_{j+1,k} + 6Q_{j,k} \\
& -4Q_{j-1,k} + Q_{j-2,k}\Big)
\end{aligned} \tag{13}
$$

with a similar definition for the transverse coordinate. This central difference scheme is very efficient and the method of choice in most cases.

For a more computationally intensive approach, a characteristic-based spatial discretization using upwind flux-differencing is included as an alternative where the explicit spatial operator may be written in terms of numerical flux vectors treated on a locally one-dimensional basis.

$$
\begin{aligned}
R(Q_{j,k}) = & \frac{1}{\Delta x}\left[\widetilde{E}_{j+\frac{1}{2},k} - \widetilde{E}_{j-\frac{1}{2},k}\right] \\
& +\frac{1}{\Delta y}\left[\widetilde{F}_{j,k+\frac{1}{2}} - \widetilde{F}_{j,k-\frac{1}{2}}\right] - H_{j,k}
\end{aligned} \tag{14}
$$

Here, the 1st order accurate numerical flux vector is defined for the streamwise coordinate by

$$
\begin{aligned}
\widetilde{E}_{j+\frac{1}{2},k}^{(1)} = \frac{1}{2}\Big[& E_{j,k} + E_{j+1,k} \\
& -|\widetilde{A}|_{j+\frac{1}{2},k}(Q_{j+1,k} - Q_{j,k})\Big]
\end{aligned} \tag{15}
$$

where the Jacobian coefficient matrix $\widetilde{A}$ is evaluated using the Roe averaged quantities[10,11]

$$
\begin{aligned}
\rho_{j+\frac{1}{2},k} &= \sqrt{\rho_{j+1,k}\rho_{j,k}} \\
u_{j+\frac{1}{2},k} &= \frac{u_{j,k}\sqrt{\rho_{j,k}} + u_{j+1,k}\sqrt{\rho_{j+1,k}}}{\sqrt{\rho_{j,k}} + \sqrt{\rho_{j+1,k}}} \\
v_{j+\frac{1}{2},k} &= \frac{v_{j,k}\sqrt{\rho_{j,k}} + v_{j+1,k}\sqrt{\rho_{j+1,k}}}{\sqrt{\rho_{j,k}} + \sqrt{\rho_{j+1,k}}} \\
h_{j+\frac{1}{2},k} &= \frac{h_{j,k}\sqrt{\rho_{j,k}} + h_{j+1,k}\sqrt{\rho_{j+1,k}}}{\sqrt{\rho_{j,k}} + \sqrt{\rho_{j+1,k}}} \\
c_{j+\frac{1}{2},k}^{2} &= (\gamma - 1)\left[h_{j+\frac{1}{2},k} - \frac{q_{j+\frac{1}{2},k}^{2}}{2}\right]
\end{aligned} \tag{16}
$$

Similar relations hold for the transverse fluxes in terms of the Jacobian coefficient matrix $\widetilde{B}$.

As developed, the above 1st order accurate scheme exhibits robust nonoscillatory behavior, but spatial resolution is poor. Direct substitution of 2nd order difference formulas does not improve matters either since this leads to the generation of spurious oscillations. To circumvent this difficulty, Harten[12] has introduced the total-variation-diminishing (TVD) property and derived sufficient conditions for constructing schemes which satisfy this property. Such constructions have been carried out for both explicit and implicit time integration schemes.[13,14] The mechanisms currently in use for satisfying TVD conditions are based on some kind of gradient limiting procedure. We adapt such a procedure based on a nonlinear flux limiter. The second-order convective flux vector is obtained by adding corrective terms to the first-order flux vector. A generalized form which may be specialized for second-order central, second-order upwind, or third-order biased upwind schemes is given by Hsieh[1] as

$$
\begin{aligned}
\widetilde{E}_{j+\frac{1}{2},k}^{(2)} = \widetilde{E}_{j+\frac{1}{2},k}^{(1)} + & \frac{1-\kappa}{4}\left[\Delta\widehat{E}_{j-\frac{1}{2},k}^{+} - \Delta\widehat{E}_{j+\frac{3}{2},k}^{-}\right] \\
& + \frac{1+\kappa}{4}\left[\Delta\widehat{E}_{j+\frac{1}{2},k}^{+} - \Delta\widehat{E}_{j+\frac{1}{2},k}^{-}\right]
\end{aligned} \tag{17}
$$

where

$$
\begin{aligned}
\Delta\widehat{E}_{j+\frac{1}{2},k}^{-} &= (MT)_{j+\frac{1}{2},k}\Delta\widehat{\sigma}_{j+\frac{1}{2},k}^{-} \\
\Delta\widehat{E}_{j+\frac{1}{2},k}^{+} &= (MT)_{j+\frac{1}{2},k}\Delta\widehat{\sigma}_{j+\frac{1}{2},k}^{+}
\end{aligned} \tag{18}
$$

$$
\begin{aligned}
\Delta\widehat{\sigma}_{j+\frac{1}{2},k}^{-} &= \mathrm{minmod}[\Delta\sigma_{j+\frac{1}{2},k}^{-}, \beta\Delta\sigma_{j-\frac{1}{2},k}^{-}] \\
\Delta\widehat{\sigma}_{j+\frac{1}{2},k}^{+} &= \mathrm{minmod}[\Delta\sigma_{j+\frac{1}{2},k}^{+}, \beta\Delta\sigma_{j+\frac{3}{2},k}^{+}]
\end{aligned} \tag{19}
$$

$$
\begin{aligned}
\Delta\sigma_{j+\frac{1}{2},k}^{-} &= \frac{1}{2}(\Lambda - |\Lambda|)_{j+\frac{1}{2},k} \\
& \quad (T^{-1}M^{-1})_{j+\frac{1}{2},k}(Q_{j+1,k} - Q_{j,k}) \\
\Delta\sigma_{j+\frac{1}{2},k}^{+} &= \frac{1}{2}(\Lambda + |\Lambda|)_{j+\frac{1}{2},k} \\
& \quad (T^{-1}M^{-1})_{j+\frac{1}{2},k}(Q_{j+1,k} - Q_{j,k})
\end{aligned} \tag{20}
$$

and the minmod operator is defined by

$$\text{minmod}[x, y] = \text{sgn}(x) \max[0, \min\{|x|, y\,\text{sgn}(x)\}] \quad (21)$$

Similar forms hold for the transverse flux vector. The values of κ corresponding to the type of scheme are presented in Table 1. The constant β is a compression parameter which is restricted to the range $1 \leq \beta \leq (3-\kappa)/(1-\kappa)$ with $\beta = 6$ when $\kappa = 1$.

Boundary Conditions

Boundary conditions for the computational analysis are implemented according to the propagation of information along the flow characteristics. On a locally one-dimensional basis, for instance, each eigenvalue may be associated with a particular characteristic. If the eigenvalue is positive, its corresponding characteristic is right/up running. Conversely, if the eigenvalue is negative, its characteristic is left/down running. When a characteristic runs out of the computational domain, the boundary condition depends on the internal flow field, and a numerical boundary condition must be formulated which depends on the computed solution. The number of unknown flow parameters minus the number of numerical boundary conditions gives the number of physical boundary conditions which must be imposed.

For the subsonic inflow boundary, there are three in-running characteristics requiring three imposed physical conditions. In this analysis, we demand that the incoming gas-phase mass flux, flow angle (swirl), and entropy remain constant. The fixed gas-phase mass flux inflow implies the injection rate is insensitive to fluctuations at the injector face. Pressure-sensitive injection may be considered by including an appropriate injector admittance condition.

For the outflow boundary, the flow is supersonic in the diverging section of the nozzle and all characteristics are out-running so that no physical conditions need be specified. However, it suffices for developmental purposes to consider a subsonic outflow boundary condition at the nozzle entrance by making the short-nozzle approximation.[15] Here, the flow is assumed to accelerate to sonic velocity in zero length. The physical condition imposed for the single in-running characteristic in this case is a fixed Mach number. Although this approximation is strictly valid for pure longitudinal-mode oscillations only, we utilize the approximation for the sake of simplicity even when transverse velocity components appear. At worse, this approach underestimates the nozzle damping effect. For the purpose of obtaining a mean flow prior to making a stability calculation, we also have the option of specifying the static pressure at the nozzle entrance.

Boundary conditions for the transverse direction are irrelevant for the annular combustor geometry in that it is only necessary to insure periodicity. When specializing for rectangular combustor geometries, solid wall boundary conditions are invoked by forcing the normal velocity component to zero.

Spray Combustion Formulation

Treatment of the propellant spray is simplified by neglecting the atomization process and assuming the injection of groups of discrete droplets at specified time intervals. Each droplet group is then represented by a probability density function (pdf) in space. The behavior of the droplets within each group is determined from the history of a single discrete computational parcel whose instantaneous position corresponds to the mean of the pdf. Conversely to the gas phase, these parcels are treated in a Lagrangian fashion. Knowing the pdf's mean location and size at a given instant, we may calculate the probable number of physical droplets in each cell to evaluate interphase transport.

The purpose of the pdf representation is to minimize the numerical shot noise which normally results when each parcel is described by a delta function point source and the droplet spray is undersampled. The numerical noise problem arising from the point source representation may be minimized by using a large number of parcels but with great sacrifice in computational efficiency. This fundamental difficulty can be so troubling that an Eulerian description of the spray is generally preferred. By introducing a distributed parcel representation, we hope to alleviate the generation of nonphysical numerical noise normally associated with a Lagrangian approach without suffering severe computational penalty.

In this development, we employ a rectangular uniform pdf shape as depicted schematically in Fig. 2. The streamwise width of this pdf corresponds to the product of the propellant injection velocity and a specified injection time interval. We currently impose an injection time interval such that this width is about 1.5 times the streamwise cell length. The width of the pdf in the transverse direction is currently equated to the transverse cell length. In addition, we require that these characteristic dimensions of the pdf shape remain fixed throughout a parcel's lifetime. A general theory has been developed to incorporate the time evolving width of the pdf due to turbulent dispersion,[16] but we are neglecting turbulence effects in the present development.

The mean position of each pdf is determined from Lagrangian tracking of a single computational parcel characterizing the behavior of the physical droplets in the group. The equations governing particle motion

are

$$\frac{d\vec{u}_p}{dt} = \frac{3}{4}\frac{\rho_g}{\rho_p}\frac{C_D}{d_p}|\vec{u}_g - \vec{u}_p|(\vec{u}_g - \vec{u}_p) \tag{22}$$

$$\frac{d\vec{x}_p}{dt} = \vec{u}_p \tag{23}$$

For the sake of efficiency, we linearize Eq. (22) by holding $(\rho_g/\rho_p)(C_D/d_p)|\vec{u}_g - \vec{u}_p|$ constant over the computational time step and integrate analytically. This gives the updated parcel velocity as

$$\vec{u}_p \approx \vec{u}_{p_0} \exp\left[\frac{-\Delta t}{\tau}\right] + \vec{u}_g\left\{1 - \exp\left[\frac{-\Delta t}{\tau}\right]\right\} \tag{24}$$

The parcel position is obtained through an even simpler treatment by using explicit Euler integration

$$\vec{x}_p \approx \vec{x}_{p_0} + \vec{u}_p \Delta t \tag{25}$$

These approximate equations yield minimal error for small computational time steps and are sufficient for engineering calculations. Evaluation of parcel motion is simplified even further in the current analysis by neglecting parcel motion in the transverse direction. This simplification may be relaxed as desired but at increased computational expense.

Physical modeling of the unsteady spray combustion process is an essential element of the stability analysis. A comprehensive description should be able to account for the effect of local flow fluctuations, in the form of organized flow oscillations as well as turbulence, on interphase transport and the propellant combustion rate. Frankly, such a complete description does not exist. Even the best current modeling techniques retain a measure of uncertainty and are generally quite complicated and encumbering. Because we wish to avoid such unnecessary complication during the developmental process, we have adapted the relatively simple d^2-law for droplet vaporization and have included simple multiplicative corrections for convective effects and pressure sensitivity. The instantaneous combustion rate of a parcel thus has the form

$$\dot{m} = \frac{\pi}{4}\rho_p d_p \beta K_v K_p \tag{26}$$

where the correction factor for convective effects is given by the Ranz-Marshall correlation as[17]

$$K_v = 1 + 0.3 Re^{1/2} Pr^{1/3} \tag{27}$$

The factor representing pressure sensitive combustion response is derived from the classical two-parameter $n - \tau$ formulation proposed by Crocco.[18] In the $n - \tau$ instability model, the propellant is assumed to undergo combustion after some time lag τ following injection. The mass burning rate during combustion is also assumed to be proportional to pressure raised to the power n. Recognizing that our Lagrangian description of the spray yields a time lag effect somewhat analogous to that of the τ parameter, we concern ourselves only with the pressure-sensitive combustion response strength as represented by the n parameter. We then note that for small pressure changes about the mean the power law pressure dependence is approximately represented by a proportional relationship between burning rate and instantaneous pressure. Thus, expressing the burning rate as the sum of a mean and fluctuating quantity $\dot{m} = \overline{\dot{m}} + \dot{m}'$, we introduce a pressure-sensitive combustion response index α, analogous to n, such that

$$\frac{\dot{m}'}{\overline{\dot{m}}} = \alpha\frac{p'}{\overline{p}} \tag{28}$$

If we further assume that the d^2-law corrected for convective effects only represents the mean burning rate, we may write the pressure-sensitive correction factor as

$$K_p = 1 + \alpha\frac{p'(t; x_p)}{\overline{p}(x_p)} \tag{29}$$

The notation expressing the functional dependency of the pressure fluctuation implies that $p'(t; x_p)$ be evaluated at time t at the instantaneous location of the parcel x_p. This approach is somewhat artificial but serves as a useful construct for developmental purposes and is not devoid of merit as a phenomenological description.

Computational Methodology

Application of the above modeling scheme requires a two-part approach. First, a mean flow must be computed by specifying the mean combustor pressure at the nozzle entrance and setting $\alpha = 0$. Because of the unsteadiness induced by the convection correction, it is generally not possible to get a fully converged steady-state flow field in this way. We avoid this difficulty by freezing the source terms once it becomes obvious that additional convergence is impossible and then continue the calculation. With the source terms frozen the calculation coverges very rapidly. For a multidimensional analysis, we also find it expedient to import a 1D mean flow solution. Once a mean flow field is obtained for reference by whatever method, stability calculations are performed by restarting the code using the short-nozzle boundary condition and a specified combustion response index.

Results and Discussion

There are two important combustor characteristics which influence stability. One is the localized combustion response to flow field fluctuations. Another is the mean combustion distribution. It is these characteristics which greatly determine the tendency for a pressure wave to be amplified or attenuated. In our combustion model, the combustion response index and, to a certain extent, the Ranz-Marshall convection correlation describe the combustion response. On the other hand, the mean spray combustion distribution is controlled by the mean burning rate constant in conjunction with the Lagrangian tracking formulation.

Based on this spray combustion model, we have used the constructed computational methodology to investigate the stability characteristics of a representative generic combustor. The propellant properties considered are given in Table 2. The combustor physical conditions are given in Table 3. Because we wished to examine the effect of combustion distribution on stability, particularly longitudinal-mode, a mean flow field was computed for three different burning rate constants to obtain a range of mean spray lengths. The burning rate constants and their resulting mean spray lengths are summarized in Table 4. The mass flow rate, Mach number, and temperature distributions for the computed mean flow fields are shown in Fig. 3.

Our approach was two prong in that we first performed a longitudinal-mode stability analysis using a pure 1D description and then examined tangential-mode instability using the equivalent annular combustor formulation. In both cases, we varied the presure sensitive combustion response index to determine the levels needed for the combustor to become intrinsically unstable. The results of this evaluation are described in the following sections.

Longitudinal-Mode Stability Characteristics

The longitudinal-mode calculations were performed using characteristic-based upwind flux differencing for spatial discretization. The 1D grid contained 101 evenly spaced nodes. To determine intrinsic combustor stability, we examined the pressure history at the nozzle entrance as it evolved following restart from the mean flow with nonzero combustion response. Triggered instability was also briefly examined by pulse combusting a specified fraction of the spray following restart.

In Fig. 4, stable combustion behavior is demonstrated for a spray length to combustor length ratio of 0.6 and a combustion response index of 0.9. Clearly there is a pressure oscillation in the combustor, but its amplitude is too small to be of significance. In contrast, Fig. 5 shows intrinsic instability when the combustion response index is increased to 1.0. Here we see classical exponential growth of the pressure wave in the linear regime followed by attainment of a nonlinear limit cycle oscillation where amplification is balanced by damping and nozzle losses. The limit cycle oscillation manifests a well defined steep-fronted pressure wave which is truly periodic. The absence of numerical noise is noteworthy.

Indeed, the spectral composition of the waveforms in Figs. 4 and 5 reveal that they are virtually noiseless. The low amplitude pressure wave observed during stable combustion is seen to be almost sinusoidal. However, the nonlinear limit cycle pressure wave observed during unstable combustion has a substantial contribution from higher harmonics. In both cases the fundamental frequency is found to correspond with the classical acoustical frequency. This behavior was characteristic of all calculations.

For additional insight, the fluctuations in velocity, temperature, and entropy are shown in relation to fluctuations in pressure during unstable combustion in Fig. 7. These waveforms are clearly periodic, but they are far from sinusoidal. The nonlinear character of the wave phenonmena during unstable combustion is evident. The pressure profile within the unstable combustor is also shown in Fig. 8 at various time intervals where we see a single pressure wave impinging the injector head and being reflected toward the nozzle.

As a summary of the longitudinal-mode stability calculations, a stability map has been constructed and presented in Fig. 9. In this map, the intrinsic stability limit is plotted against a normalized combustion response strength and a normalized mean spray length. The combustion response strength is defined as the product of the response index and the mean burning rate since it is this product which determines the magnitude of the fluctuating burn rate as the spray length is expanded or contracted. This is evident from an inspection of Eqs. (26) and (29). What we find is that as the spray combustion region is expanded to encompass a larger portion of the combustor, damping effects increase. While some energy is always added in phase with the wave, with an increasingly larger combustion region there is an increase in energy added out of phase with the wave. The net effect is greater stability. In contrast, as the spray combustion region is contracted and concentrated near the injector face, there is a simultaneous reduction in the out-of-phase contributions and increase in the in-phase contributions. The net effect is greater instability.

To make an important point concerning our instability model, the pressure history for a combustion pulsed (3% of total spray) stability calculation is pre-

sented in Fig. 10. Here, we find the same stability behavior as for the non-pulsed case except that the development of the limit cycle occurs much faster. The resulting waveforms are identical to that obtained for non-pulsed stability. This is further demonstrated by the spectral composition of the waveform in Fig. 11. The reason for this behavior is that the combustion response is linearly related to pressure as seen in Eq. (28). For a nonlinear model, pulsed stability could differ greatly from non-pulsed stability.

Tangential-Mode Stability Characteristics

All tangential-mode calculations were performed using 6th order central differencing for spatial discretization. The 2D grid for the annular combustor geometry contained 51 nodes in the axial coordinate and 30 nodes in the tangential coordinate. The transverse length was chosen to match the first tangential mode, and the annular width was adjusted to obtain the correct cross-sectional area. The inflow was constrained to have zero swirl. Again, intrinsic stability behavior was examined by restarting from a mean flow field imported from the 1D analysis.

Pressure histories for a spray length to combustor length ratio of 0.6 and a combustion response index of 0.1 are given in Fig. 12 demonstrating the development of an intrinsic tangential-mode instability. These pressures were recorded at 1/10 of the combustor length from the injector and at 3 tangential locations differing by a phase angle of $\pi/2$. We observe the evolution of an extremely high-amplitude pressure wave traveling around the combustor although combustion response is extremely small. In fact, instability was observed for this combustor even with zero pressure sensitive combustion response. Evidently, the Ranz-Marshall convective correlation offers enough sensitivity to velocity fluctuations to support a pressure wave. The reason for this behavior is attributed to minimal damping from drag, streamwise convective losses, and out-of-phase energy contributions. Obviously, the combustor is much more susceptible to tangential-mode instabilities than longitudinal-modes in accord with practical experience.

Also observable is mode switching which occurs during the 10 to 25 msec time interval following restart. The short term instability up to 10 msec consists of a single unidirectional traveling high-amplitude pressure wave. This is evidenced by an exploded view of the short term pressure histories in Fig. 13. We clearly see that the unstable pressure waveforms at $\theta = 0$ and $\theta = \pi$ are exactly half a period out of phase. In the exploded view, we also observe that the pressure wave is almost entirely positive biased, is shocklike in nature, and has a fundamental frequency corresponding to the first spinning tangential-mode acoustical frequency of the chamber. Furthermore, there are some superimposed high frequency components which show up spectrally in Fig. 14 consisting of low level noise and possible precursors to mode switching.

Following mode switching, the pressure waveforms at $\theta = 0$ and $\theta = \pi$ are exactly in phase and oscillate at twice the fundamental frequency. This long term instability mode is shown in Fig. 15 during the time interval 35 to 40 msec after restart. Here, we also see that the pressure at $\theta = \pi/2$ is now oscillating at the fundamental frequency with very high amplitude peaks centered between the lower amplitude ones manifested at $\theta = 0$ and $\theta = \pi$. Obviously, the instability now consists of two pressure waves traveling in opposing directions which meet at $\theta = \pi/2$ and $\theta = 3\pi/2$ to form two extremely large pressure spikes on opposite sides of the combustor.

The pressure field behavior during instability is exhibited more clearly in the flooded contour plots of Figs. 16 and 17 which show the combustor pressure history during the short term and long term modes, respectively. A revealing feature discernable from these flooded contour plots is the observation that the severity of the instability is localized near the injector face. The magnitude of the pressure wave diminishes rapidly downstream most probably as a result of increased streamwise convective losses as the Mach number increases. The wave is also curved with the nozzle end leading the injector end. One can clearly ascertain the rationale and effectiveness of baffles in controlling tangential-mode instabilities from these results.

Another noteworthy feature of this analysis is that the basic features of the instability are in accord with experimental results for a combustor with similar physical conditions.[19] The research combustor used in the referenced experimental work was 11 inches in diameter while the combustor of this analysis is about 10 inches in diameter. Experimentally, a single high-amplitude shocklike pressure wave was observed sweeping about the chamber at the injector face with a period of 470 μs corresponding to the first spinning tangential-mode acoustical frequency. The pressure wave was found to decay very rapidly in amplitude and curve in the direction of rotation as one moved away from the injector toward the nozzle entrance. For the short term instability of our calculations, the period was about 435 μs in correspondence with the acoustical frequency, and we clearly have a strong shocklike pressure wave decaying rapidly in strength and curving in the direction of rotation downstream from the injector. That these essential features are common is encouraging and indicative that the analysis has indeed captured at least the basic physics.

Conclusion

Based on preliminary evaluation, the constructed computational methodology appears promising as a research tool and potential design/development aide for investigating the stability characteristics of liquid rocket engines. The low numerical noise properties of the method are very favorable, and Lagrangian treatment of the spray offers improved flexibility for direct modeling of spray combustion. In our judgment, the achievement of low noise while using a Lagrangian technique is noteworthy considering that the number of computational parcels was kept to a minimum through the use of probability distributions for the propellant spray. The qualitative agreement between a predicted tangential-mode instability, where a shocklike pressure wave was found to sweep about the chamber near the injector face, with existing experimental results is also encouraging.

Although the current development has been confined to a simplified linear pressure-sensitive combustion response model, more comprehensive physical submodeling may easily be implemented as desired. In fact, the most important use of the developed computational tool may well be as a research test bed for investigating the effect of various physical submodels on combustor stability characteristics. In conjunction with experimental data, much can be learned about the physical nature of combustion driven instabilities. With time, as modeling becomes more refined through an interplay between experiment and computation, it should be possible to evolve a support capability for aiding in design/development and investigating new combustor concepts.

Acknowledgments

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Nomenclature

English Symbols

A, B	Jacobian coefficient matrices
$\widetilde{A}$, $\widetilde{B}$	Roe-averaged Jacobian coefficient matrices
b	annular combustor width (m)
C_p	constant-pressure specific heat $(J/kg\,K)$
C_v	constant-volume specific heat $(J/kg\,K)$
C_D	drag coefficient,
	$= 24(1 + Re^{\frac{2}{3}}/6)/Re \;\; ; Re < 1000$
	$= 0.44 \;\; ; Re \geq 1000$
c	speed of sound (m/s)
d	diameter (m)
$\vec{\mathcal{D}}$	momentum exchange per unit volume (N/m^3),
	$= \widetilde{n}\frac{\pi}{8}d_p^2\rho C_D\lvert\vec{u}_p - \vec{u}_g\rvert(\vec{u}_p - \vec{u}_g)$
E, F	convective flux vectors
$\widetilde{E}$, $\widetilde{F}$	numerical flux vectors
$\widehat{E}$, $\widehat{F}$	gradient limited flux vectors
e	total internal energy (J/kg)
g	gravitational acceleration (m/s^2)
H	source vector
h	specific stagnation enthalpy (J/kg)
K_v	Ranz-Marshall forced-convection correction
K_p	pressure-sensitive combustion rate correction
l	transverse combustor length (m)
l_c	streamwise combustor length (m)
M	conservative to primitive variable transformation matrix or mach number
$\dot{m}$	mass vaporization rate (kg/s)
$\widetilde{n}$	particle density (particles/m^3)
Pr	Prandtl number
p	pressure (Pa)
Q	conserved variable vector
q	modulus of velocity (m/s)
Δq	net specific energy generation (J/kg)
R	ideal gas constant $(J/kg\,K)$ or explicit spatial difference operator
Re	particle diameter based Reynolds number
s	specific entropy $(J/kg\,K)$
T	similarity transformation matrix or temperature (K)
t	time (s)
u, v	velocity components (m/s)
V	primitive variable vector, $= b[\rho, u, v, p]^T$
$\dot{w}$	mass vaporization rate per unit volume $(kg/m^3\,s)$
x, y	space coordinates (m)

Greek Symbols

α	presure-sensitive combustion response index
β	compression parameter or burning rate constant (m^2/s)
γ	specific heat ratio
ϵ	dissipation parameter
κ	spatial discretization selection parameter
Λ	eigenvalue matrix
λ	conductivity (W/mK)
μ	viscosity (kg/ms)
ρ	density (kg/m^3)
σ	flux-limiter parameter
τ	particle relaxation time (s),
	$= (4/3)(\rho_p/\rho_g)(d_p/C_D)(1/\lvert\vec{u}_g - \vec{u}_p\rvert)$

Subscripts and Superscripts

g	gas-phase property

i	grid index, parcel group index
j	grid index
l	liquid spray
n	time index
o	original value
p	parcel property
x, y	coordinate components
$+$	downstream influence
$-$	upstream influence
$*$	reference condition
$\vec{()}$	vector
$\overline{()}$	mean value
$()'$	fluctuation about mean

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Table 1 Scheme Switching Parameter

Scheme	κ
2nd Order Upwind	-1
2nd Order Central	1
3rd Order Biased Upwind	1/3

Table 2 Propellant Properties

γ	1.2
C_p	1800 J /kg K
Δq	$20(10)^6$ J / kg
ρ_l	750 kg / m^3
μ	$5(10)^{-5}$ kg / m s
λ	$1(10)^{-1}$ W / m K

Table 3 Combustor Physical Conditions

l_c	0.5 m
A_c	0.05 m^2
$\overline{P}_n$	34 atm
$\dot{m}_g$	40 kg / s
$\dot{m}_l / \dot{m}_g$	0.25
d_0	100 μm
$u_{l\,0}$	75 m / s

Table 4 Mean Burning Rate / Spray Length

β	l_e / l_c
1.4	0.36
1.0	0.60
0.8	0.90

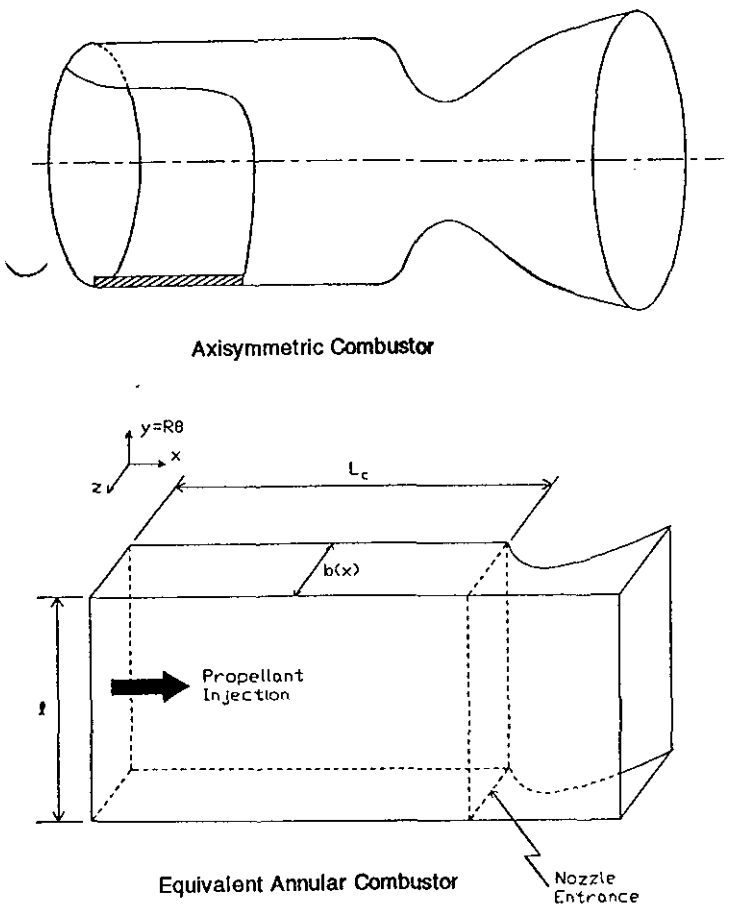

Figure 1. Schematic of physical combustor and equivalent annular computational geometry.

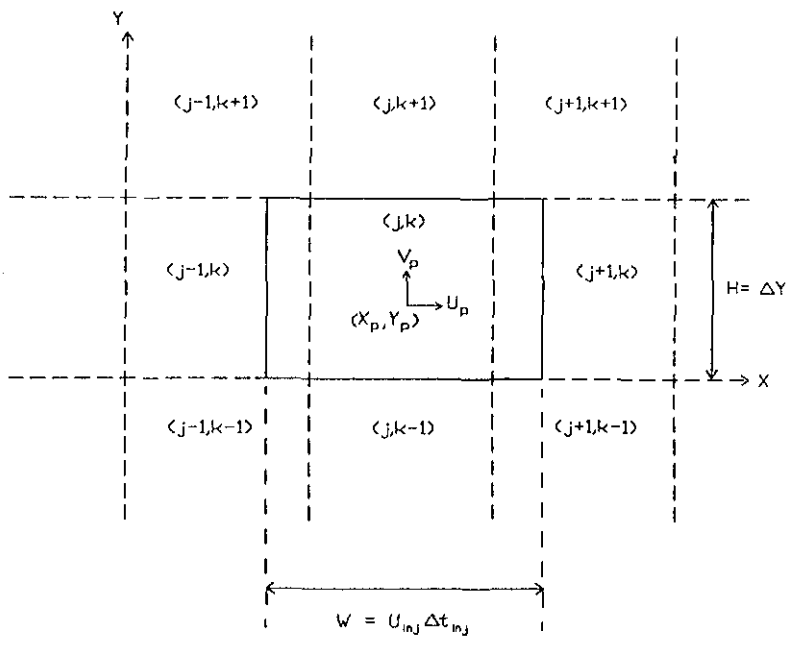

Figure 2. Schematic of rectangular uniform pdf in relation to computational grid.

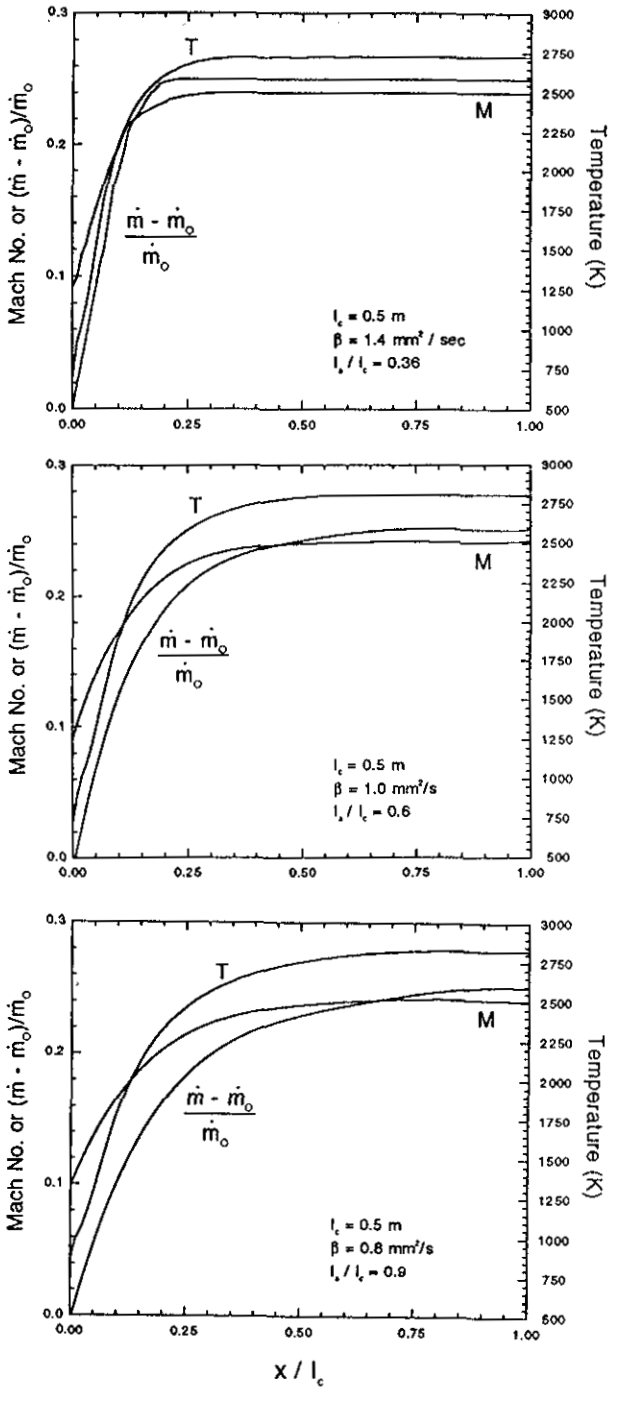

Figure 3. Computed mean flow field.

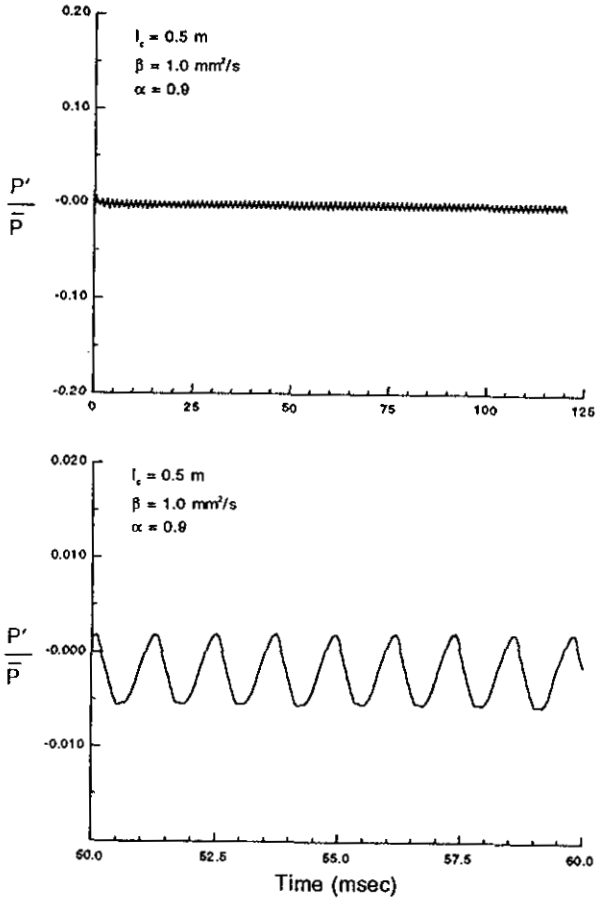

Figure 4. Longitudinal-mode, non-pulsed combustor stability calculation (stable).

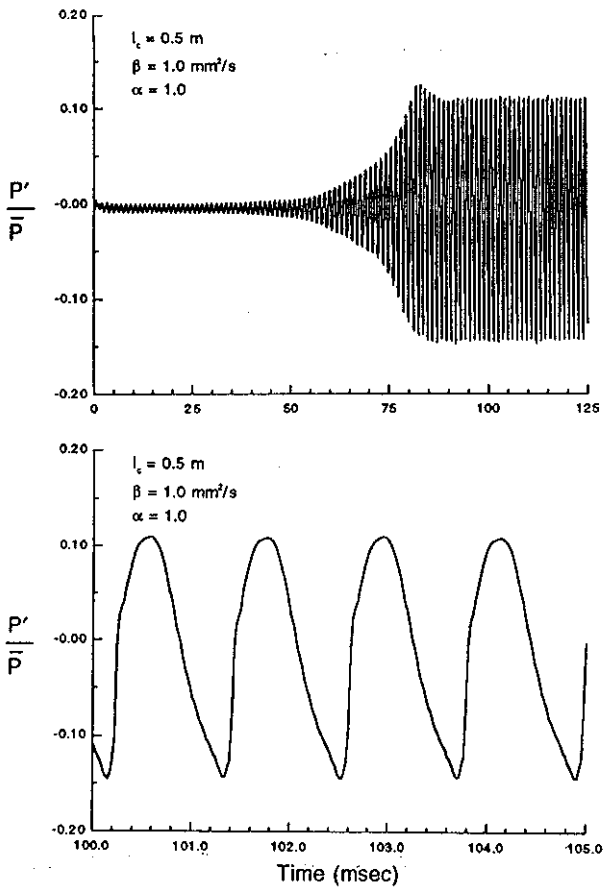

Figure 5. Longitudinal-mode, non-pulsed combustor stability calculation (unstable).

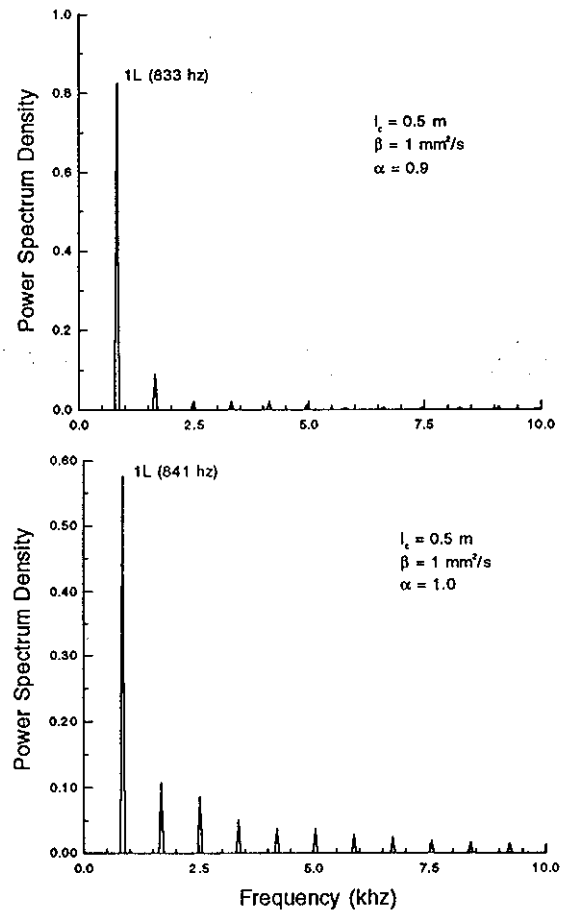

Figure 6. Spectral composition of pressure wave during stable and unstable combustion.

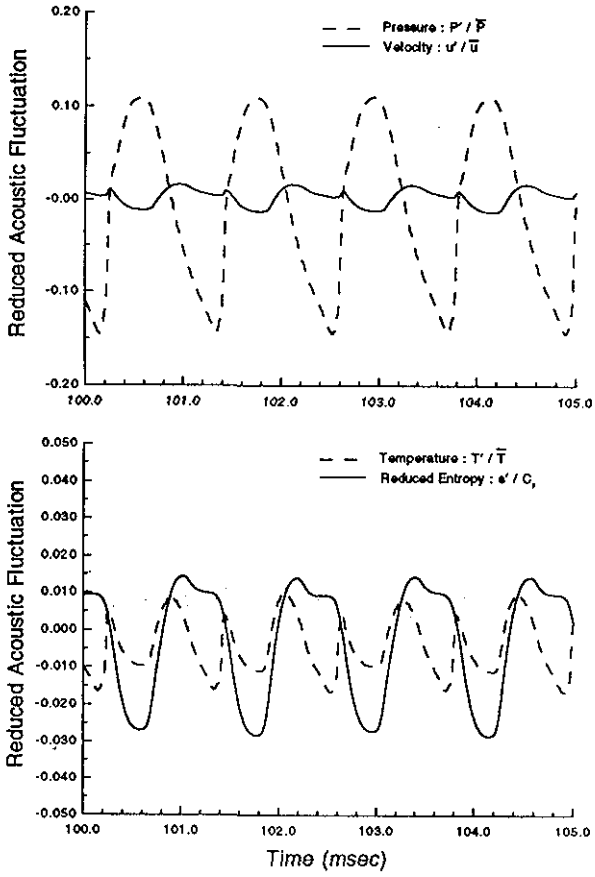

Figure 7. Fluctuation of flow parameters during unstable combustion.

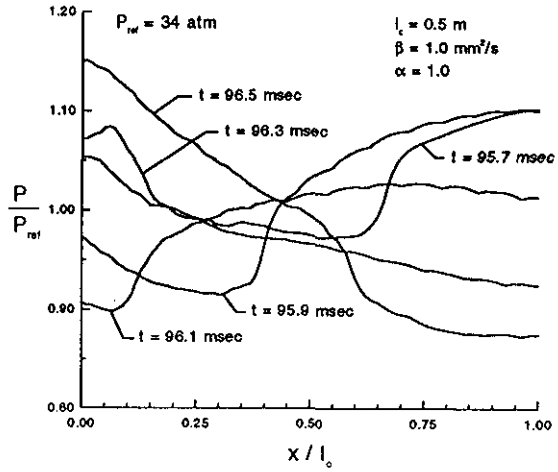

Figure 8. Pressure profiles during unstable combustion.

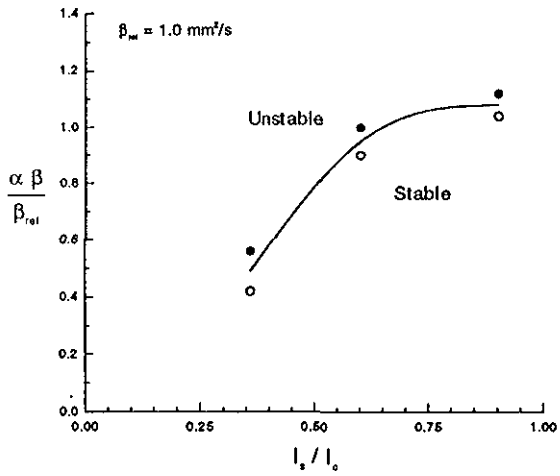

Figure 9. Longitudinal-mode combustor stability map.

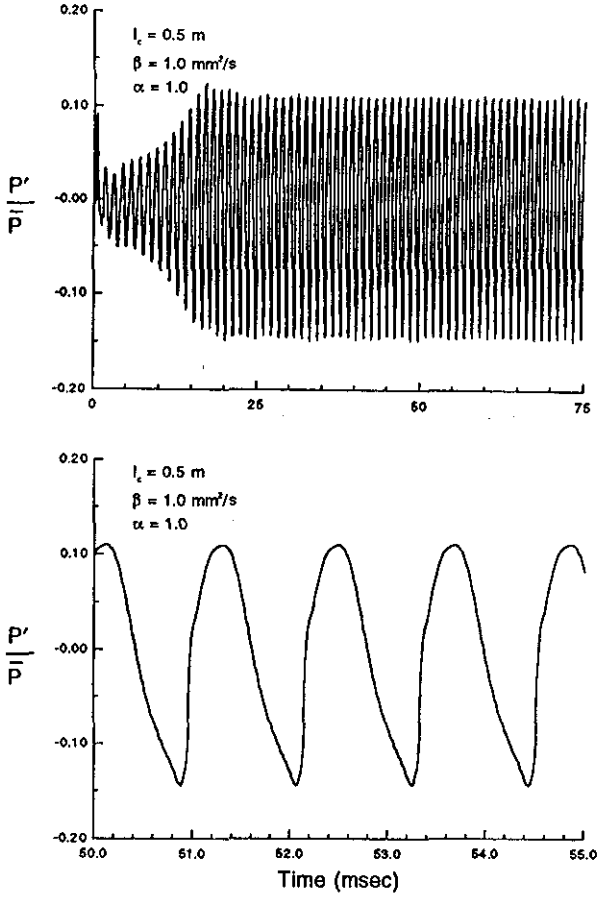

Figure 10. Longitudinal-mode, pulsed combustor stability calculation (unstable).

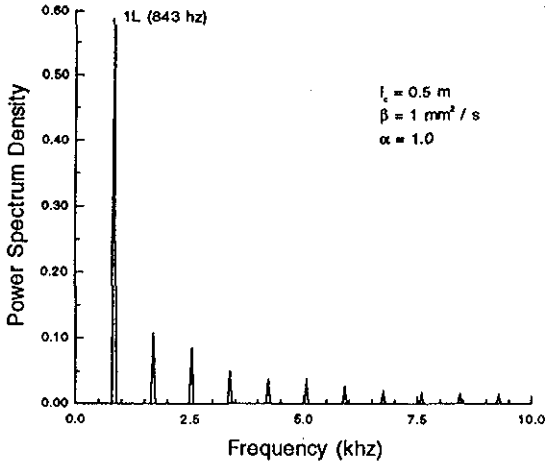

Figure 11. Spectral composition of pressure wave from pulsed stability calculation.

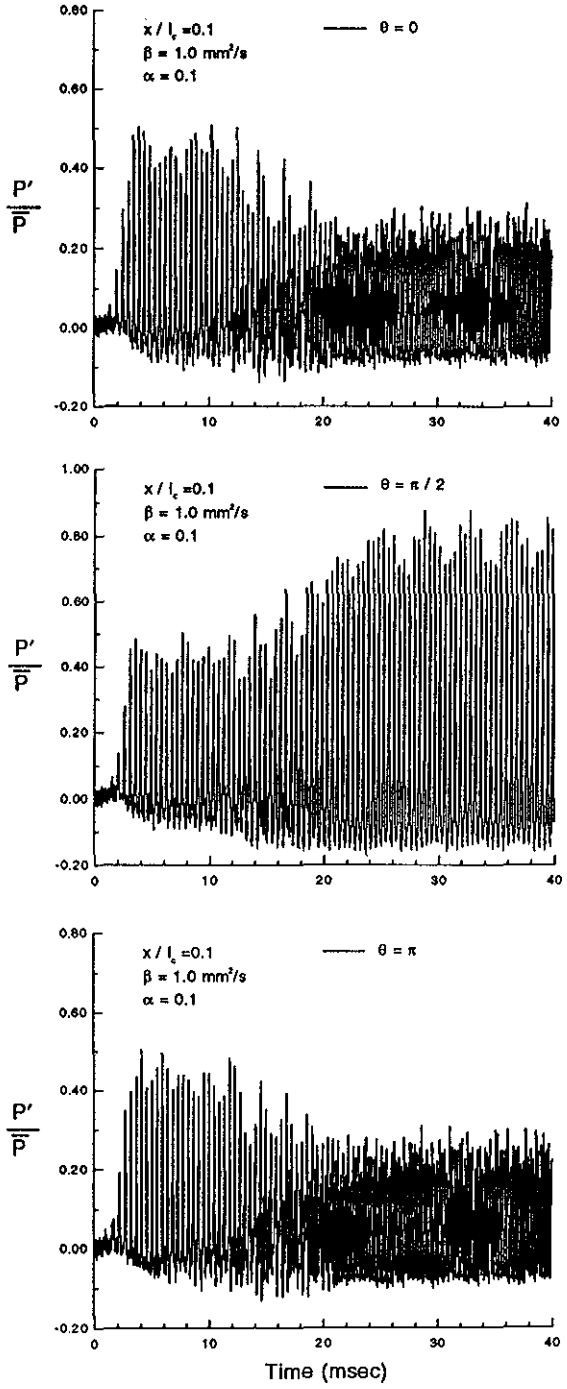

Figure 12. Tangential-mode, non-pulsed combustor stability calculation (unstable).

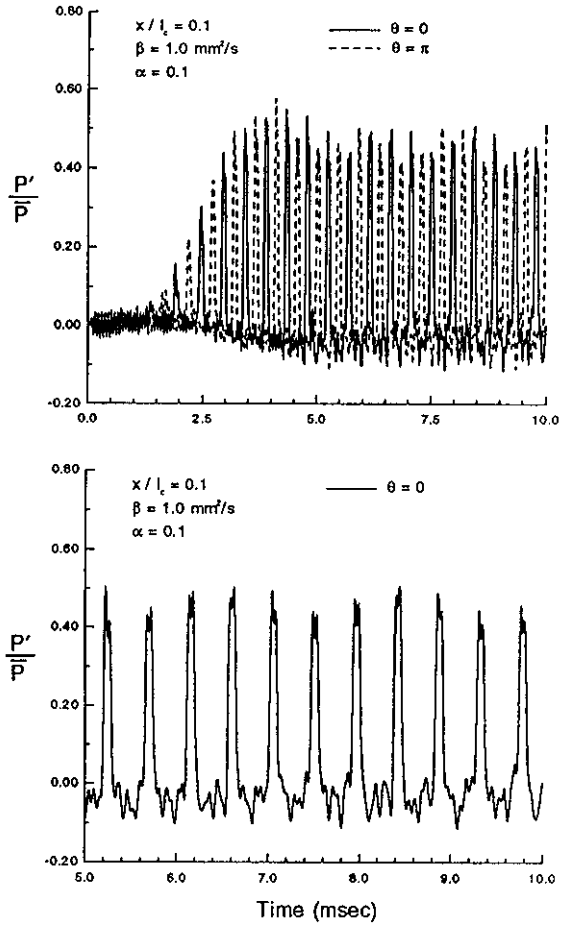

Figure 13. Short term tangential-mode instability.

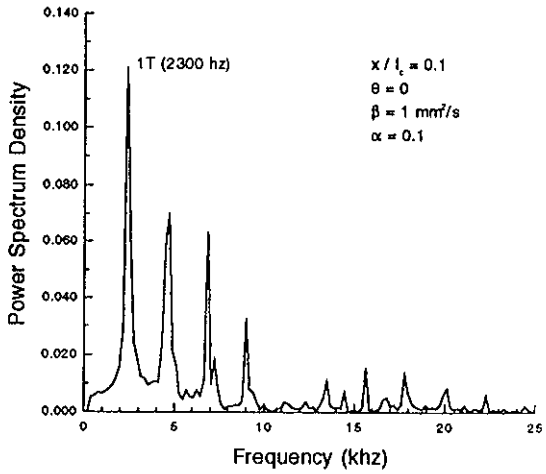

Figure 14. Spectral composition of tangential pressure wave for short term mode.

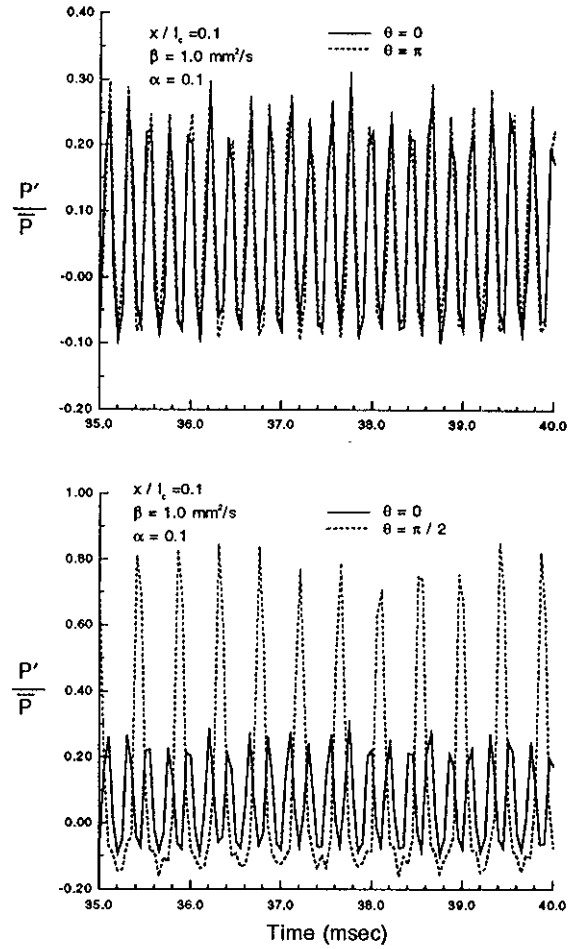

Figure 15. Long term tangential-mode instability.

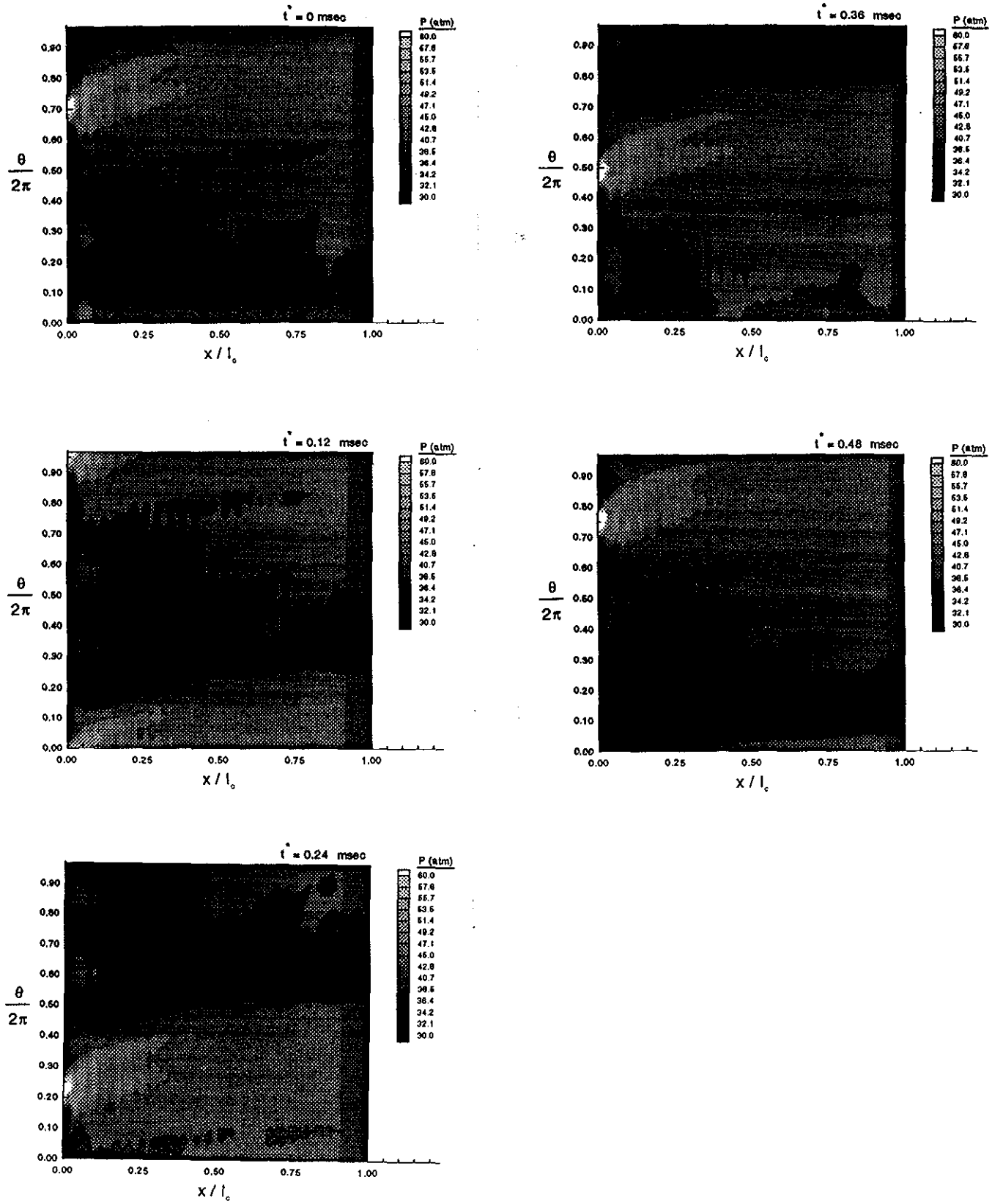

Figure 16. Combustor pressure history over one period of short term mode.

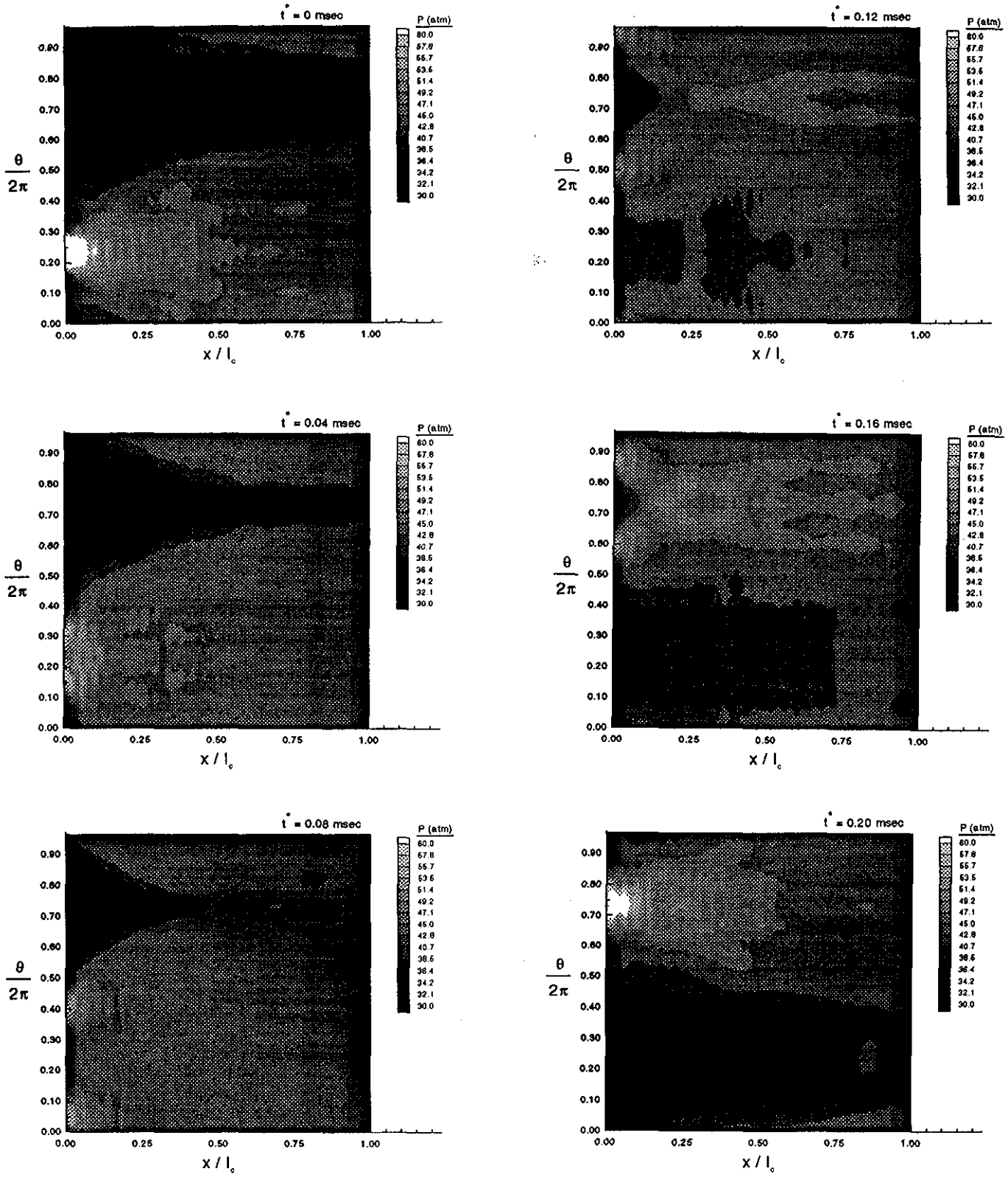

Figure 17. Combustor pressure history over one-half period of long term mode.